Table S1. The 162 environmentally relevant antibiotics screened in this study, with therapeutic family, abbreviation used throughout the manuscript, and the database from which the structure was retrieved.

#	Antibiotic	Family	Abbreviation	Source
1	Sulfadiazine	Sulfonamides	SDZ	DrugBank
2	Sulfapyridine		SPD	DrugBank
3	Sulfathiazole		STZ	DrugBank
4	Trimethoprim		TMP	DrugBank
5	Sulfamerazine		SMR	DrugBank
6	Sulfamethazine (sulfadimidine)		SMZ	DrugBank
7	Sulfamethoxazole		SMX	DrugBank
8	Sulfamethizole		SMT	DrugBank
9	Sulfamethoxypyridazine		SMP	DrugBank
10	Sulfameter		SFM	DrugBank
11	Sulfaguanidine		SGD	DrugBank
12	Sulfacetamide		SCT	DrugBank
13	Sulfachloropyridazine		SCP	DrugBank
14	Sulfadoxine		SDX	DrugBank
15	Sulfamonomethoxine		SMM	ChEMBL
16	Sulfaquinoxaline		SQX	DrugBank
17	Sulfadimethoxypyrimidine		SDT	PubChem
18	Sulfadimethoxine		SDM	DrugBank
19	Sulfisomidine		SSM	ChEMBL
20	Sulfisoxazole		SSZ	DrugBank
21	Sulfanilamide		SNM	DrugBank
22	N-acetyl sulfamethoxazole		NSM	PubChem
23	N-acetyl sulfadiazine		NSD	PubChem
24	N-acetyl sulfapyridine		NSP	PubChem
25	N-acetyl sulfamethazine		NSZ	PubChem
26	Sulfanitran		SNT	DrugBank
27	Benzenesulfonamide		BSN	ChEMBL
28	Dapsone		DPS	DrugBank
29	N-acetyl sulfadimethoxine		NSX	PubChem
30	N-acetyl sulfathiazole		NST	PubChem
31	N-acetyl sulfamerazine		NSN	PubChem
32	Ormetoprim		OMP	ChEMBL
33	Sulfabenzamide		SBZ	DrugBank
34	Sulfamoxole		SMO	DrugBank
35	Sulfaphenazole		SPZ	DrugBank
36	Sulfasalazine		SSL	DrugBank
37	Oxytetracycline	Tetracyclines	OTC	DrugBank
38	Tetracycline		TTC	DrugBank
39	Chlorotetracycline		CTC	ChEMBL
40	Doxycycline		DXC	DrugBank
41	Isochlortetracycline		ICC	PubChem
42	Epitetracycline		ETC	ChEMBL
43	4-Epianhydrochlortetracycline		ECC	ChEMBL
44	Minocycline		MIN	DrugBank
45	Anhydrotetracycline		ATC	ChEMBL
46	Methacycline		MTC	DrugBank
47	4-Epianhydrotetracycline		EHT	ChEMBL
48	4-Epichlortetracycline		ETT	ChEMBL
49	4-Epioxytetracycline		EXT	PubChem
50	Anhydrochlortetracycline		ACT	PubChem
51	Demeclocycline		DMC	DrugBank
52	Meclocycline		MCC	DrugBank
53	Rolitetracycline		RTC	DrugBank
54	β-apo-oxytetracycline		AOT	PubChem
55	Ofloxacin	Quinolones	OFX	DrugBank
56	Norfloxacin		NFX	DrugBank
57	Ciprofloxacin		CFX	DrugBank
58	Enrofloxacin		EFX	DrugBank

59	Flumequine		FMQ	DrugBank
60	Sparfloxacin		SFX	DrugBank
61	Gatifloxacin		GFX	DrugBank
62	Fleroxacin		FLX	DrugBank
63	Lomefloxacin		LFX	DrugBank
64	Enoxacin		ENX	DrugBank
65	Danofloxacin		DNX	DrugBank
66	Difloxacin		DFX	DrugBank
67	Marbofloxacin		MBX	DrugBank
68	Moxifloxacin		MFX	DrugBank
69	Pefloxacin		PFX	DrugBank
70	Sarafloxacin		SRX	DrugBank
71	Nalidixic acid		NXA	DrugBank
72	Oxolinic acid		OXA	ChEMBL
73	Levofloxacin		LVX	DrugBank
74	Pipemidic Acid		PPA	ChEMBL
75	Piromidic Acid		PMA	ChEMBL
76	Cinoxacin		CNX	DrugBank
77	Clinafloxacin		CLX	DrugBank
78	Nadifloxacin		NDX	DrugBank
79	Orbifloxacin		OBX	DrugBank
80	Tosufloxacin		TFX	DrugBank
81	Erythromycin	Macrolides	ETM	ChEMBL
82	Anhydroerythromycin		HTM	ChEMBL
83	Roxithromycin		RTM	ChEMBL
84	Clarithromycin		CTM	ChEMBL
85	Azithromycin		ATM	DrugBank
86	Spiramycin		SPM	DrugBank
87	Josamycin		JSM	DrugBank
88	Tylosin		TYS	ChEMBL
89	Tilmicosin		(TM1-TM2)	ChEMBL
90	Midecamycin		MCM	ChEMBL
91	Descladinosyl azithromycin		DAM	ChEMBL
92	N-Demethyl azithromycin		(NM1-NM2)	ChEMBL
93	N'-Desmethyl azithromycin		(NS1-NS2)	ChEMBL
94	Descladinosyl clarithromycin		DCM	ChEMBL
95	Erythromycin A enol ether		EAE	ChEMBL
96	Erythromycin A 6,9-imino ether		EAI	PubChem
97	Erythromycin A oxime		EAO	PubChem
98	Erythromycylamine		EMC	ChEMBL
99	Neospiramycin I		NPM	PubChem
100	Oleandomycin		ODM	ChEMBL
101	Phosphorylated azithromycin		PAM	PubChem
102	Amoxicillin	β - Lactams	AXC	DrugBank
103	Ampicillin		APC	ChEMBL
104	Cephalexin		CPX	ChEMBL
105	Cefotaxime		CTX	ChEMBL
106	Penicillin G		PNG	ChEMBL
107	Cephradine		CPD	ChEMBL
108	Penicillin V		PNV	ChEMBL
109	Cefoperazone		CPZ	ChEMBL
110	Cefuroxime		CFM	ChEMBL
111	Cefadroxil		CDX	ChEMBL
112	Aztreonam		AZT	ChEMBL
113	Cefaclor		CFC	ChEMBL
114	Cefapirin		CFP	PubChem
115	Cefazolin		CFZ	ChEMBL
116	Cefdinir		CFN	ChEMBL
117	Cefetamet pivoxyl		CMP	DrugBank
118	Cefixime		CFI	ChEMBL
119	Cefoxitin		CFT	ChEMBL
120	Cefquinome		CFQ	ChEMBL

121	Ceftazidime		CTZ	ChEMBL
122	Ceftiofur		CFF	ChEMBL
123	Ceftriaxone		CFR	ChEMBL
124	Cloxacillin		CXC	ChEMBL
125	Dicloxacillin		DCX	ChEMBL
126	Meropenem		MPM	ChEMBL
127	Nafcillin		NFC	ChEMBL
128	Oxacillin		OXC	ChEMBL
129	Piperacillin		PPC	ChEMBL
130	Amikacin	Aminoglycosides	AKC	ChEMBL
131	Streptomycin	Aminoglycosides	STM	ChEMBL
132	Spectinomycin	Aminoglycosides	SCM	ChEMBL
133	Thiamphenicol	Amphenicols	TPN	ChEMBL
134	Chloramphenicol		CPN	ChEMBL
135	Florfenicol		FPN	ChEMBL
136	Narasin	Polyethers	NRN	ChEMBL
137	Salinomycin		SLM	ChEMBL
138	Lasalocid		LSC	ChEMBL
139	Monensin		MNN	ChEMBL
140	Lincomycin	Lincosamides	LCM	ChEMBL
141	Clindamycin	Lincosamides	CDM	ChEMBL
142	Metronidazole	Nitroimidazoles	MTZ	ChEMBL
143	Dimetridazole		DMZ	ChEMBL
144	Ornidazole		ODZ	ChEMBL
145	Ronidazole		RDZ	ChEMBL
146	Virginiamycin M1	Streptogramins	VGM	ChEMBL
147	Furazolidone	Nitrofurans	FZD	ChEMBL
148	Furaltadone		FTD	ChEMBL
149	Nifuroxazide		NFZ	ChEMBL
150	Nitrofurantoin		NFT	ChEMBL
151	Tiamulin	Diterpenoids	TML	ChEMBL
152	Bacitracin	Polypeptides	BTC	PubChem
153	Vancomycin	Glycopeptides	VMC	ChEMBL
154	Rifampicin	Rifamycins	RFP	ChEMBL
155	Rifaximine		RFX	ChEMBL
156	Fusidic acid	Others	FSA	ChEMBL
157	2-Quinoxalinecarboxylic acid		QXA	ChEMBL
158	Novobiocin		NVB	ChEMBL
159	Griseofulvin		GOF	ChEMBL
160	Fumagillin		FMG	ChEMBL
161	Fluconazole		FCZ	ChEMBL

Table S2. Spearman correlation coefficients (p) between physicochemical descriptors and mean docking energy for each alginate–chitosan nanoparticle formulation.

Molecular Descriptor	1-1	1-2	2-1
Heavy atoms	-0.65	-0.67	-0.73
Molecular weight	-0.63	-0.65	-0.71
Hydrogen-bond acceptors	-0.54	-0.55	-0.61
Topological polar surface area	-0.55	-0.56	-0.60
Hydrogen-bond donors	-0.51	-0.51	-0.57
Formal charge	0.42	0.42	0.41
Fraction sp3	-0.26	-0.28	-0.35
Rotatable bonds	0.00	-0.03	-0.05
LogP	0.18	0.17	0.14
Aromatic rings	-0.03	-0.02	0.03

Table S3. Nanoparticle structural stability and polymer interaction network for the three formulations. Radii are the 95th percentile of polymer heavy-atom distances from the polymer center of mass; radial compaction is the ratio of the final to the initial radius, and the apparent volume ratio is its cube. Final values are averaged over the last 20 ns of simulated time. Hydrogen-bond and salt-bridge counts describe the polymer's own internal network, isolated as the frame-wise minimum across the antibiotics sharing each trajectory and normalized by the number of chains of the relevant type. Values are mean ± one standard deviation across three replicas.

Property	1:1	1:2	2:1
Alginate chains	30	30	60
Chitosan chains	30	60	30
Ca^{2+} ions	120	120	240
Initial radius P95 (Å)	52.2	56.2	52.9
Final radius P95 (Å)	66.3 ± 7.4	131.4 ± 3.9	53.0 ± 1.1
Radial compaction	1.27 ± 0.14	2.34 ± 0.07	1.00 ± 0.02
Apparent volume ratio	2.10 ± 0.69	12.84 ± 1.08	1.01 ± 0.06
Final Rg (Å)	41.2 ± 2.9	62.1 ± 2.6	39.3 ± 0.4
Initial SASA (10^3 Å^2)	125	186	135
Final SASA (10^3 Å^2)	103 ± 1	179 ± 5	126 ± 3
Shape anisotropy	0.025	0.02	0.005
Alginate–alginate H-bonds per alginate chain	2.85 ± 0.27	3.20 ± 0.13	4.31 ± 0.16
Alginate–chitosan H-bonds per chitosan chain	3.42 ± 0.25	2.41 ± 0.03	5.61 ± 0.43
Ca^{2+} salt bridges per frame	61 ± 4	88 ± 4	100 ± 5

Table S4. Molecular dynamics adsorption metrics for each antibiotic–nanoparticle combination. Values are mean ± one standard deviation across three independent 100 ns replicas; each system contained eight copies of every antibiotic. A molecule counts as in contact when at least one of its heavy atoms lies within 3.5 Å of a polymer heavy atom, and as stably bound when it satisfies that criterion for more than half of the final 20 ns. Median residence is the median duration of a contact event. Excess H-bonds are the antibiotic-specific contribution obtained after subtracting the frame-wise minimum across the antibiotics sharing a trajectory. Antibiotic codes: ATC, anhydrotetracycline; MTC, methacycline; CPZ, cefoperazone; VGM, virginiamycin M1; VCM, vancomycin; RFP, rifampicin.

Alg:Chi	Antibiotic	Molecules contacted (of 8)	Contact time, whole run (%)	Contact time, final 20 ns (%)	Stably bound (of 8)	Contact events (run^{-1})	Median residence (ns)	Longest contact (ns)	Events >10 ns	Excess H-bonds (frame^{-1})	D (Å^2 ns^{-1})
1:1	ATC	6.0 ± 1.0	9.4 ± 6.5	12.7 ± 7.3	0.7 ± 0.6	27 ± 10	1.08	20.2 ± 15.7	2.0	0.45 ± 0.32	20.9 ± 8.2
1:1	MTC	4.7 ± 2.1	11.2 ± 7.8	8.1 ± 6.8	0.7 ± 0.6	15 ± 9	1.49	33.8 ± 21.7	3.0	0.97 ± 0.64	19.3 ± 0.7
1:1	CPZ	6.7 ± 1.5	11.7 ± 9.1	16.9 ± 14.8	1.3 ± 1.2	28 ± 19	1.16	31.2 ± 30.4	2.0	0.92 ± 0.80	32.3 ± 22.6
1:2	ATC	8.0 ± 0.0	26.9 ± 11.5	30.5 ± 21.5	2.3 ± 1.5	76 ± 7	0.99	28.9 ± 22.8	4.7	1.25 ± 1.07	24.1 ± 6.2
1:2	MTC	8.0 ± 0.0	18.2 ± 4.6	21.5 ± 4.9	1.3 ± 0.6	61 ± 16	1.07	26.1 ± 8.4	3.0	0.51 ± 0.23	15.6 ± 3.9
1:2	CPZ	8.0 ± 0.0	27.9 ± 3.7	29.6 ± 12.0	2.3 ± 1.2	55 ± 11	1.15	58.6 ± 19.3	5.0	2.64 ± 0.34	21.0 ± 7.4
2:1	ATC	3.7 ± 2.3	11.1 ± 11.1	12.9 ± 17.8	1.0 ± 1.7	21 ± 19	1.99	32.5 ± 32.5	1.3	0.84 ± 0.97	34.1 ± 30.7
2:1	VGM	3.0 ± 0.0	2.5 ± 1.0	4.2 ± 6.7	0.3 ± 0.6	8 ± 1	0.66	14.8 ± 7.6	0.7	0.19 ± 0.02	22.4 ± 8.1
2:1	VCM	3.3 ± 1.5	8.1 ± 6.5	11.5 ± 12.4	1.0 ± 1.0	12 ± 9	4.06	33.7 ± 31.2	2.0	0.20 ± 0.19	36.4 ± 4.2
2:1	RFP	2.0 ± 1.7	0.8 ± 0.8	0.5 ± 0.5	0.0 ± 0.0	4 ± 4	0.41	2.7 ± 3.0	0.0	0.03 ± 0.03	34.6 ± 5.8

Table S5. Comparison of the radius-referenced positional state assignment with the direct-contact criterion. State percentages are the mean occupancy over the whole trajectory and three replicas. The discrepancy column is the difference between the two definitions of the bound fraction, in percentage points; it is small for the structurally stable 1:1 and 2:1 particles and large for the 1:2 particle, whose expanding boundary overtakes stationary antibiotic molecules. The final column reports how often that happens explicitly, as the percentage of frames classified AB-SORBED in which no polymer atom lies within 3.5 Å, reported per antibiotic; pooled over the three antibiotics of the 1:2 system the value is 61.8% (Figure S4). Antibiotic codes as in Table S3.

Alg: Chi	Anti-biotic	FREE (%)	ADSORBED (%)	ABSORBED (%)	Radius-based bound (%)	Contact-based bound (%)	Discrepancy (pp)	ABSORBED frames with no contact (%)
1:1	ATC	84.9	9.9	5.1	15.1	9.4	5.6	13.7
1:1	MTC	84.7	7.2	8.1	15.3	11.2	4.1	5.9
1:1	CPZ	83.1	11.9	5.0	16.9	11.7	5.2	7.9
1:2	ATC	12.9	35.6	51.5	87.1	26.9	60.2	58.9
1:2	MTC	12.7	38.5	48.8	87.3	18.2	69.2	70.3
1:2	CPZ	13.0	40.6	46.4	87.0	27.9	59.2	55.9
2:1	ATC	87.6	12.4	0.0	12.4	11.1	1.2	0.0
2:1	VGM	95.8	3.3	1.0	4.2	2.5	1.7	6.4
2:1	VCM	90.6	9.4	0.0	9.4	8.1	1.3	0.0
2:1	RFP	98.2	1.8	0.0	1.8	0.8	1.0	0.0